

■ Electro, Physical & Theoretical Chemistry

Electrochemical Exfoliation of Graphite to Fluorographene: An Effect of Degree of Functionalization on $2\text{Br}^-/\text{Br}_2$ Redox ReactionMunaiah Yeddala,^{*,[a]} Tharangattu N. Narayanan,^[b] Ragupathy Pitchai,^{*,[a]} and Vijayamohanan K. Pillai^[a]

In this work, we employed the electrochemical exfoliation (EE) to synthesize and tune the degree of functionalization in fluorinated graphene (FG) and studied its (degree of functionalization) effect on bromine redox reaction. The degree of functionalization was tuned by varying the concentration of the electrolyte. Physical characterization of as-synthesized FG reveals that fluorine and oxygen content varies from 2.32 to 3.92 atomic % and from 10.46 to 21.16 atomic % respectively. The concentration of fluoride to hydroxide ion ratio should be at an optimum level in the electrolyte to get maximum degree of fluorination, which shows a sizeable effect on bromine redox

reaction. The bromine redox reaction is more reversible on highly FG and less reversible on less FG and Oxygen functionalities has no effect on the reaction. The kinetic parameters such as rate constant (k) (varying from 1.45×10^{-3} to $52.43 \times 10^{-3} \text{ mA/cm}^2$) and exchange current density (i_0) (varying from 2.61×10^{-4} to $9.92 \times 10^{-4} \text{ cm/s}$) suggest that nature and degree of functionalization play a vital role to improve the kinetics of redox reaction of bromine. This investigation can pave paths to design new and sustainable electrode materials for bromine-based redox flow batteries.

Introduction

Among all the carbon materials, graphene is one of the well-studied analogues for a decade due to its spectacular ability to tune chemical, physical and electronic properties by functionalization.^[1–11] Doping of heteroatom such as nitrogen,^[12,13] phosphorous,^[14] sulfur,^[15–17] boron^[18,19] and halogens^[20] which can open the band gap can further change drastically the properties of the material. Unlike other heteroatoms such as boron, nitrogen, phosphorous and sulfur, halogen forms a single bond with carbon atom^[21] which allows understanding the structure and properties of the functionalized graphene. Among all Halogenated graphenes (HG), fluorinated graphene is more stable^[22] and due to its high electronegativity, fluorine can tune the band gap so that the electrochemical properties improve effectively compared to than that of other elements.^[23] For instance, the Band gap of the FG depends on the degree of fluorination.^[23–25] Moreover, FG is one of the most useful materials in diverse applications ranging from biological,^[26,27] energy harvesting^[28] and storage,^[2,28] sensing,^[29] to electronic applications.^[30,31] Electrochemical properties of the fluorinated graphene sheets can

also be tuned by varying the fluorine content attached to the graphene sheet. For instance, Praveen Meduri et al.^[2] have synthesized FG with different degree of fluorination (CF_x where $x=0.47, 0.66$ and 0.89) and studied their electrochemical properties by using them as anode material in primary lithium batteries. The $\text{CF}_{0.47}$ shows superior performance over high current rates (5 C) where other materials ($\text{CF}_{0.66}$ and $\text{CF}_{0.89}$) are unable to function effectively. Similarly, Veronika et al.^[29] has synthesized FGs with different coverage in order to study the electron transfer ability to sense biomolecules and Zhao et al.^[23] tuned the fluorine content in FGs by solution phase exfoliation for studying the effect of extent of fluorine on electrochemical properties. However, many basic questions like the mechanism of fluorination, active centers for electrocatalysis in FG, the role of fluorine in improving the electrocatalytic properties, etc. are not yet clear.

Due to its exceptional properties and applications, FGs have been synthesized by using a variety of techniques and method.^[23,32–39] Table S1 shows a comparison of different methods for the synthesis of FG. Most of these involve graphene oxide (GO) as a graphene source and different fluorinated agents as a fluorine source.^[21] For instances, Wang et al.^[33] used GO and F_2 gas at 180°C to prepare FG, whereas Gong et al.^[38] used photochemical synthesis to prepare FG from GO and HF. Another method is exfoliation of graphite fluoride to fluorinated graphene oxide^[40] which resembles the hummer's method, which is well known for the synthesis of graphene oxide.^[41] Micromechanical cleavage of fluorographene has also attempted, but it is difficult to get high-quantities by using this method.^[42] However, the synthesis of FGs is labour intensive which involves several steps and

[a] Dr. M. Yeddala, Dr. R. Pitchai, Prof. V. K. Pillai
Electrochemical Power Sources Division, Fuel Cells Section Central
Electrochemical Research Institute, Karaikudi-630 003, India
E-mail: munaiah.y@gmail.com
ragupathyp@cecni.res.in

[b] Dr. T. N. Narayanan
Tata Institute of Fundamental Research, Hyderabad, Telangana 500107,
India

Supporting information for this article is available on the WWW under
<https://doi.org/10.1002/slct.201902114>

hazardous chemicals and finally, it is also difficult to control the degree of functionalization as the quality of the graphene changes dramatically due to the lack of control over the reaction parameters. Accordingly, the synthesis of high-quality FGs with the desired degree of fluorination is still a challenge.

Electrochemical exfoliation (EE) is one of the important methods to synthesize functionalized graphene as it offers precise control over the degree of functionalization^[43] in addition to room temperature operation to ensure morphological invariance. For instance, Bruna et al.^[44] synthesized graphene oxyfluoride by using electrochemical exfoliation of graphite in an aqueous solution of HF acid. However, this method could not tune the degree of functionalization and the quality of graphene is indeed poor. In our earlier work^[45] we synthesized HG through EE in an aqueous solution of potassium halide, utilized them as an improved electrode material in zinc bromine redox flow battery (ZBB). Among all HG, FG shows superior performance as bromine electrode in ZBB in terms of energy efficiency and durability for 350 cycles. Since the FG shows good performance as a bromine electrode material in ZBB, we intended to study the effect of fluorine content on bromine redox reaction. In this work, accordingly, we propose an eco-friendly synthesis of the FGs with different degree of functionalization by systematically changing the concentration of the electrolyte (KF aqueous solution). Bromine redox couple based redox flow batteries (RFB) including ZBB need materials not only to sustain for a long time but also to enhance the kinetics of bromine redox reaction. We strongly believe that the present study not only helps to understand the effect of extent of fluorination on bromine redox reaction but to design the sustainable electrode materials for bromine electrode also in bromine-based redox flow batteries.

Results and Discussion

Electrochemically synthesized FGs are analyzed using different physical characterization techniques. Accordingly, figure 1 shows the comparative X-ray diffraction (XRD) pattern of graphite and FGs synthesized using an aqueous solution of KF having different concentrations. The XRD pattern of FGs revealing a sharp peak at 26° (2 θ) corresponding to (002) plane of graphite is strong evidence for the formation of few layers of graphene without much change in the crystal lattice. As shown in the inset, a small shift in the peak position is evidence for the exfoliation of the graphene layers^[45,46] perhaps due to intercalation of species. Interestingly, the intensity of peak corresponding to the plane (002) changes drastically than that of graphite powder due to a reduction in the number of graphene layers.^[46] The peak intensity decreases and FWHM of the peak increase gradually from FG–M4 to FG–M1, and the peak intensity corresponds to a number of layers and lateral size of the graphene. The changes (see table 1) in the peak (due to change in the number of layers and FWHM is due to change in the size of the graphene) may be due to the difference in the quantity of hydroxide and fluoride ions at the anode during the exfoliation. From the above results, it is confirmed that a low concentration of KF solution leads to less

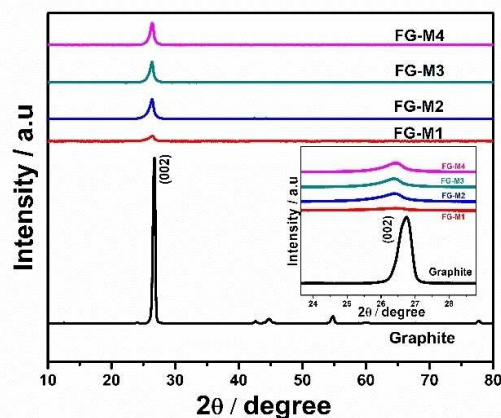


Figure 1. Comparative XRD patterns of FGs synthesized through EE and Graphite. The concomitant changes in the peak position and intensity are strong evidence for the change in the size and number of layers of graphene sheets in the FGs.

Table 1. XRD data obtained from Figure 2. The change in Peak position, d-spacing and FWHM indicating the substantial changes in FGs.

	2 θ / degree	FWHM	d-Spacing (Å)
Graphite	26.69	0.087	3.337
FG–M1	26.26	1.200	3.389
FG–M2	26.28	0.955	3.384
FG–M3	26.31	0.801	3.383
FG–M4	26.37	0.745	3.374

number of layers and smaller particle size in the resultant FGs. These results are in good agreement with microscopic analysis data (FESEM and TEM).

X-ray photoelectron spectroscopy (XPS) is employed to study the elemental composition of the FGs. Thus figure 2 shows a comparative XP survey spectrum of FGs with different degree of functionalization. The presence of peaks at 285, 532 and 687 eV corresponding to C1s, O1s, and F1s respectively is the primary evidence for the formation of functionalized graphene. The area under the peak in the survey spectrum is directly related to the extent of an element in the compound under investigation. As shown in figure 2, the intensities of O1s and C1s peaks do change from FG–M1 to FG–M4 which is due to the change in the degree of functionalization with the concentration of the electrolyte. The elemental compositions are calculated from the survey spectra of FGs by dividing the area under the curve with an atomic sensitive factor of the corresponding element and the data are given in table 2. Interestingly, the quantity of oxygen decreases from 21.1 to 10.6 atomic % while the carbon content increases concomitantly from 76.53 to 88.66 atomic % with a concentration of the electrolyte. However, the extent of fluorine unexpectedly increases with concentration, from 2.3 to 3.9 followed by a reduction to 2.6 atomic %. This could be attributed to the reactivity and availability of the fluoride and hydroxide ions simultaneously at the anode during exfoliation. Hence, it is

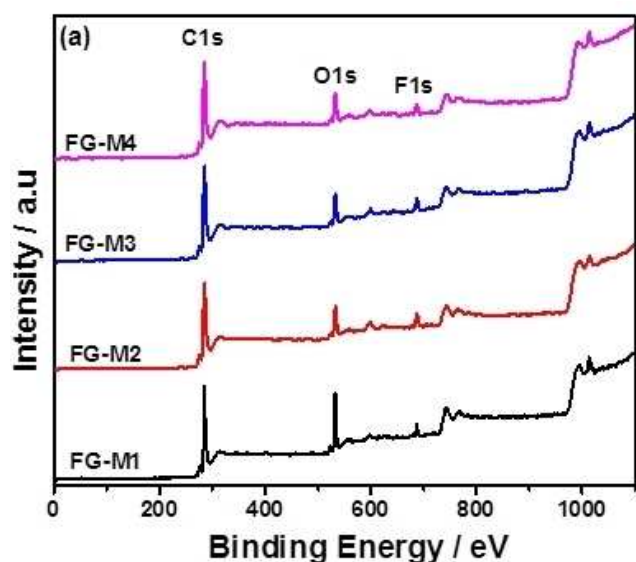


Figure 2. XPS survey spectra of FGs synthesized through EE in 1, 2, 3 and 4 M KF electrolyte solutions. The change in the intensity of the peaks corresponding to C1s O1s and F1s is a strong evidence for the formation of high-quality FGs with different degree of functionalization.

Table 2. Elemental composition of FGs calculated from survey spectrum of XPS (Figure 3).

	Atomic % of "C"	Atomic % of "O"	Atomic % of "F"
FG-M1	76.53	21.16	2.32
FG-M2	78.33	17.75	3.92
FG-M3	85.32	11.33	3.34
FG-M4	88.66	10.67	2.67

clear that with increasing concentration of KF the amount of OH^- may decrease at the electrode, which is responsible for the reduction of oxygen content in the FGs. This is applicable to the change in the atomic % of carbon too. The reason for the different degree of fluorination is, the availability and the activity of active species at the electrode (it is explained in detail in the mechanism of exfoliation). The above results help to conclude that the optimum concentration of electrolyte in order to get the maximum degree of fluorination is 2 M.

Figure 3 shows high resolution or core XP spectra of C1s, O1s and F1s. The peaks are deconvoluted into different signals in order to identify the type of bonds present as per standard protocols using a Shirley background correction algorithm. Accordingly, Figure 3a, d, g and j show the C1s of FGs. All the spectra show the peaks at 284.9 ± 0.1 , 285.9 ± 0.1 , 287.3 ± 0.1 , and 289.8 ± 0.1 eV corresponding to sp^2 , sp^3 , C–O and C=O carbon respectively.^[47] The broadening of the peak, however, decreases with the concentration of the electrolyte (from FG-M1 to FG-M4), which is due to the change in the degree of functionalization. O1s spectra deconvoluted into two peaks at 533.5 ± 0.1 and 532.3 ± 0.1 eV corresponding to oxygen bonded with carbon through double bond and single bond respectively. Core spectrum of F1s is deconvoluted into two

peaks at 687.8 ± 0.1 eV and 686.8 ± 0.2 eV corresponding to covalently bonded fluorine, semi-ionically bonded fluorine respectively.^[48] With increasing fluorine the quantity of covalently bonded fluorine becomes more than that of semi-ionically bonded fluorine (see core XP spectra of F1s).^[27]

Raman spectroscopy is one of the well-established tools to analyze the crystal structure of carbon material, in particular, graphene-based materials.^[49,50] Figure 4 accordingly shows a comparative Raman spectrum of FGs with graphite (graphite powder). All the spectra show typical peaks at 1353 cm^{-1} , 1595 cm^{-1} and 2714 cm^{-1} corresponding to D, G, and 2D bands respectively. The D band is also known as a defective band and G band is known as a graphitic band. The intensity of the D band is the measure of the density of the defect and the D band (at 1353 cm^{-1}) is more intensive in Raman spectra of FGs than that of graphite, which is due to the formation of the defects. The Intensity of the D band is found to be increased from FG-M1 to FG-M2 and decreased to FG-M4. These changes are directly related to the changes in the degree of functionalization of FGs.

As mentioned earlier (in XPS data), a higher degree of fluorination yields covalently bonded fluorine (to carbon) and the covalently bonded carbon (C–F bonded carbons) is responsible for more intensive D band in GF-M2. The G band at $1586 \pm 5 \text{ cm}^{-1}$ is also sensitive to the extent of functionalization, the slight blue shift (see table 3) in the G band is also direct evidence for functionalization. Interestingly, the intensity of the 2D band (overtone of D band) which arises due to the internal transition at K point is less intense in FGs than in graphite. This can be attributed to the damage to the crystalline structure due to functionalization.^[51] Other defective bands such as D' and G' also exhibit a similar trend of D band. I_D/I_G ratio, which is considered as a measurement for the quality of graphene, increase to FG-M2 first followed by a further reduction. By comparing the XPS results with I_D/I_G ratio, one can conclude that the FG with highest fluorine content has more defects. Interestingly I_{2D}/I_G ratio also increases from FG-M1 to FG-M4 perhaps due to reduction in the degree of functionalization and also perhaps due to an increase in the lateral size of the graphene.

The comparative FTIR spectra (See Figure S1) of graphite and FGs show peaks at 1214 cm^{-1} and 1080 cm^{-1} corresponding to C–F covalent and semi-ionic^[27,38] bonds respectively. The peaks at 3430 cm^{-1} and 1577 cm^{-1} corresponding to O–H stretch and C–O stretch respectively, are evidence for the formation of oxygen functionalities during the exfoliation. The intensity of peak corresponding to O–H stretching at 3430 cm^{-1} decreases while the peaks corresponding to C–F covalent and semi ionic bonds gradually emerge from FG-M1 to FG-M4. This could be attributed to the reduction in the degree of oxygen functionality which indirectly suggests a change in the quantity and nature of the C–F bond. These results are in excellent agreement with the elemental composition data obtained from XPS survey spectra.

Changes in the morphological features associated with the different degree of functionalization have been studied by using FESEM and TEM. Accordingly, figure 5 shows a compar-

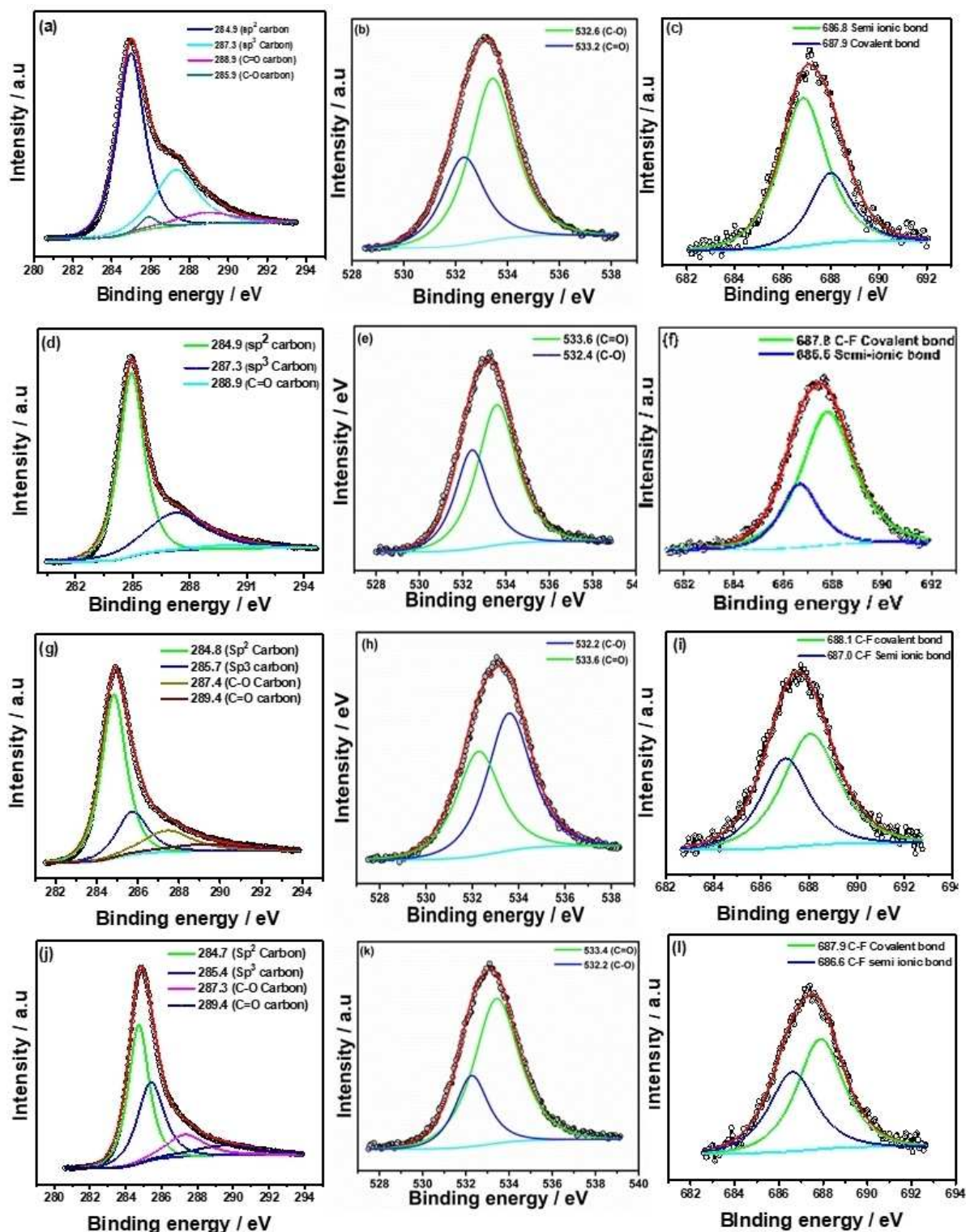


Figure 3. Core XP spectra of C1s, O1s, and F1s of FG synthesized through EE in 1 M KF (a-c), 2 M KF (d-f), 3 M KF (g-i) and 4 M KF aqueous solution. All the spectra are deconvoluted and assigned according to their binding energy

ison of the FESEM images of FGs at low (Figure 5a, 5b, 5c, and 5d) and high magnifications (Figure 5e, 5f, 5g and 5i). All FGs show few layers of graphene with exfoliated edges. FESEM images at high magnification show FGs with few layers of

graphene sheets having a smooth surface. The lateral size of the graphene sheets varies from 1 μ to 10 μ although it is slightly higher compared to FGs synthesized using other methods. Figure 6 shows the TEM images of FGs having a few

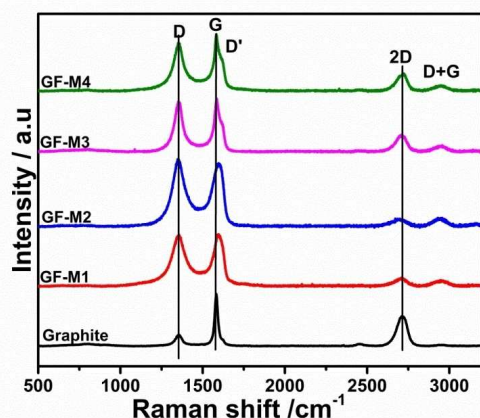


Figure 4. Raman spectra of FGs and graphite. The variation in G and D bands is strong evidence for the different degree of functionalization.

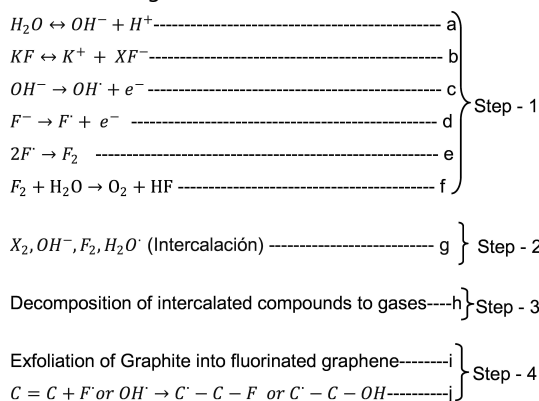
Table 3. A comparison of the main features from Raman spectra (figure 4).

	D (cm ⁻¹)	G (cm ⁻¹)	2D (cm ⁻¹)	I _D /I _G	I _{2D} /I _G
FG-M1	1353	1595	2699	0.993	0.160
FG-M2	1354	1591	2707	1.062	0.114
FG-M3	1354	1587	2709	0.937	0.301
FG-M4	1355	1583	2720	0.844	0.312
Graphite	1354	1581	2714	0.209	0.412

layers of graphene with a smooth surface and a larger lateral size. The smooth surface of the FGs is evidence for the production of high-quality graphene with fewer defects.

The mechanism of exfoliation of graphite in the KX aqueous solution is explained in detail in our earlier publication.^[45] However, certain additional features are necessary to explain, the reasons for the change in the degree of functionalization and other concomitant changes. The steps involved in the exfoliation are shown below wherein; both $OH^\bullet$ radicals and $F^\bullet$ radicals are generated from OH^- and F^- ions respectively at

the surface of the anode during the exfoliation. These radicals, being active species, their amount at the interface is squarely responsible for the change in the degree of functionalization of FGs. At a lower concentration (1 M of KF) of the electrolyte, the quantity of OH^- ions is relatively higher than that of F^- ions, and consequently, the concentration of initiators ($OH^\bullet$) of the C–C bond breaking is higher, facilitating a faster exfoliation of graphite. These $OH^\bullet$ radicals are not only responsible for C–C bond breaking but also



for the extent of oxygen content in the FGs. With increasing concentration of the electrolyte, the concentration of OH^- ions may decrease and the concentration of F^- ions may increase at the interface. Less availability of OH^- ions ($OH^\bullet$ radicals) leads to a reduction in the degree of functionalization and an increase in the time required to complete the exfoliation (see figure 7). Interestingly, XP spectroscopy also confirms the increase in the extent of fluorine content with the concentration of the electrolyte up to 2 M (concentration of the KF solution), followed thereafter by a decrease. At the optimum concentration of the electrolyte (2 M) both the active species are at optimum levels to produce maximum fluorination. The high quantity of OH^- ions not only reduces the time for exfoliation of graphite rod but also reduces the particle size (due to rapid exfoliation) of the FGs. The chronoamperometry (figure 7) and electrochemical capacitance studies (figure 8) further support our speculation about the exfoliation and functionalization.

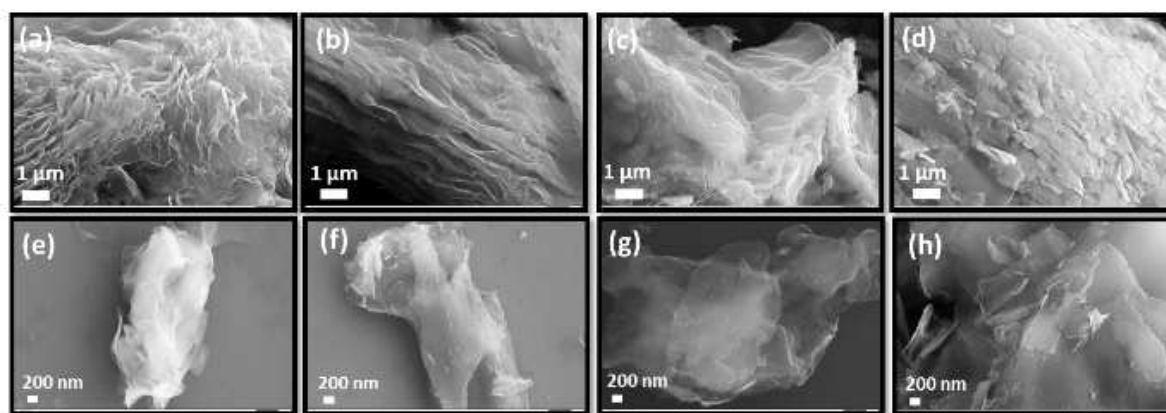


Figure 5. FESEM images of FGs at low (a,b,c,d) and high magnifications (e,f,g,h). Exfoliated edges and smooth surface can be seen in all FGs. Scale bar is 1 μm and 200 nm for low (a-d) and high (e-h) magnification images respectively.

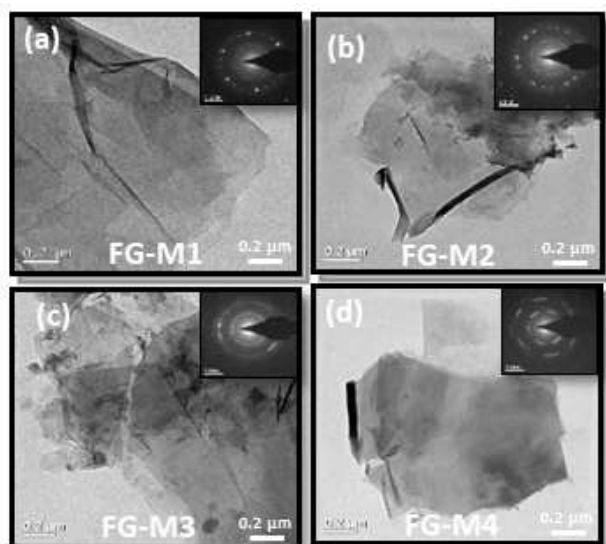


Figure 6. TEM images of FGs and corresponding selected area diffraction (SAED) shown as inserts. The defect-free surface is evidence for the formation of high-quality graphene. The scale bar is 0.2 μm .

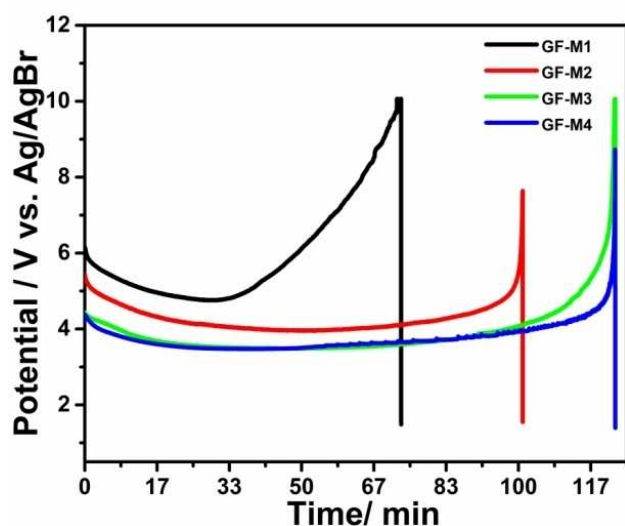


Figure 7. The chronopotentiometry curves measured during the EE of graphite in KF electrolyte having different concentrations. The time required to complete the process is directly proportional to the concentration of the electrolyte used for the EE.

Figure 7 shows chronopotentiometric curves measured using a conventional three-electrode system in KF aqueous electrolyte. A constant current of 0.2 A/cm^2 was applied till the graphite rod (anode) consumed completely. As shown in the initially the potential is found to decrease with time followed by an increase. The reduction in potential may be due to the formation of fluorine functionalization on the surface of the electrode and subsequent reduction in the surface area could gradually increase the potential (to maintain the constant current). Potential drop at the end is due to the complete

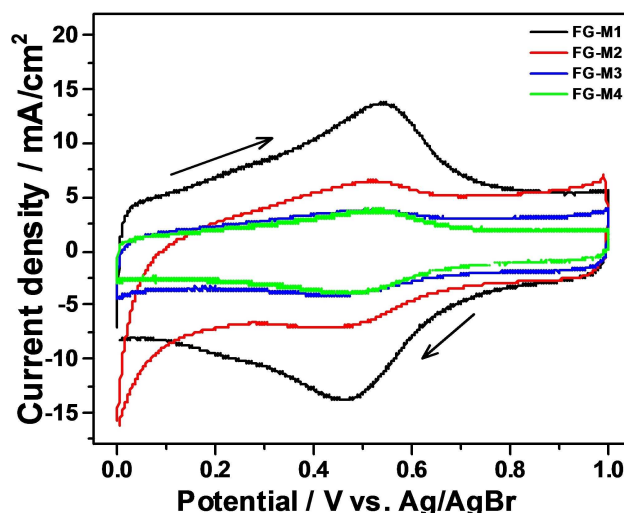


Figure 8. Cyclic voltammetry curves of FGs in 1 M H_2SO_4 electrolyte at 100 mV/s scan rate. The changes in the area of CV are related to the surface area of the electrocatalyst.

consumption of the graphite rod. The time required for the consumption of graphite rod is less in 1 M KF solution and it is increasing with concentration. The variation in exfoliation time which is good evidence for the above-proposed mechanism and Variation in the potential (initial potential where exfoliation occurs) can be attributed to the change in the concentration of the electrolyte. Figure 8 shows the CV of FGs recorded in 1 M H_2SO_4 at 100 mV/s scan rate, where the area under the curve is observed to decrease from FG-M1 to FG-M4. Moreover, the specific capacitance values calculated from CV are found to be 1.15, 0.68, 0.41 and 0.33 F/cm^2 for FG-M1, FG-M2, FG-M3 and FG-M4 respectively. The change in the specific capacitance can be attributed to variation in the surface area of the electrode. The result confirms that the FG-M1 has a higher surface area than that of other FGs which provides strong evidence for the above proposed mechanism. A sharp redox peak in the CV corresponds to the redox reaction of oxygen functionalities present in the FGs.

As synthesized FGs with different degree of functionalization are employed as an electrocatalyst for the bromine redox reaction. Accordingly, figure 9 shows the CVs of bromine redox reaction over the surface of the FGs. The changes in the CVs (expanded region shows the changes in the anodic and cathodic peaks) of $2\text{Br}^-/\text{Br}_2$ redox couple over different FGs are strong evidence for the effect of functionalization on electrocatalysis of $2\text{Br}^-/\text{Br}_2$ redox reaction. The CVs show two regions i.e. non-faradaic region (0 to $\sim 0.9 \text{ V}$) and faradaic region (0.9 to 1.4 V). The capacitive behavior of FGs is as similar as data shown in figure 8. The capacitive current is higher for FG-M1 while it decreases to FG-M4, in support of the electrochemical surface area results (figure 8). The faradaic region shows sharp peaks at 1.1 V and 0.89 V corresponding to bromide ion oxidation and bromine reduction respectively. The peak position and peak current density and onset potential are the essential parameters to be considered to evaluate the perform-

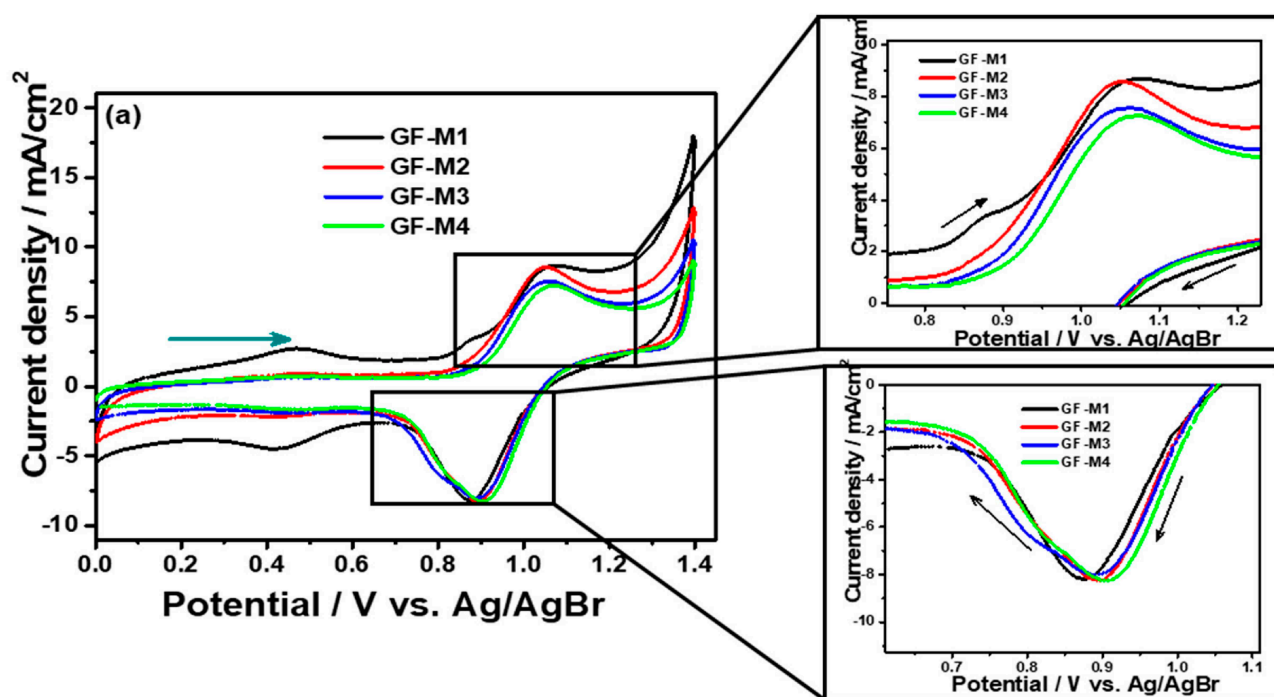


Figure 9. Cyclic voltammograms of bromine redox reaction over the surface of the FGs. The concomitant changes in the capacitive and faradaic regions showing the difference in the catalytic activity of FGs with different de

ance of an electrocatalyst. The CV data obtained from figure 9 is shown in table 4, and the peak current values are given after subtracting the capacitive current.

Table 4. Cyclic voltammetry data obtained from figure 9

	Ipa	Ipc	Ipa/Ipc	Epa	Epc	ΔE_p	E1/2	Onset
FG-M1	0.479	0.450	1.06		0.886	206	0.989	811
FG-M2	0.549	0.542	1.01	1.055	0.883	172	0.969	778
FG-M3	0.488	0.576	0.84	1.085	0.901	184	0.993	795
FG-M4	0.463	0.612	0.75	1.077	0.897	180	0.987	803

Interestingly the peak current values increase while the peak separation decrease with an increase in the fluorine content in the FGs. Accordingly, FG-M2 shows the highest current and least peak separation than that of other FGs. This states that the quantity of fluorine may play a vital role in the bromine redox reaction. Higher peak current and less peak separation make GF-M2 better electrocatalyst for the bromine redox reaction. Onset potential value is also less for FG-M2 than that of the rest, which may give additional confirmation about the fluorine active sites. The ratio of Anodic peak (Ipa) and Cathodic peak (Ipc) current is one of the good evidence for the reversibility of the redox couple on the electrode. Unlike CNT based electrodes^{52, 53} the Ipa/Ipc ration decreases from FG-M1 to FG-M4 which is perhaps due to strong adsorption of bromine over the surface the FGs having less degree of functionalization.⁵⁴ The selective adsorption of bromine redox

couple over the FGs need to be investigated in detail for better understanding.

In order to understand the nature of the electrochemical reaction, CV is recorded at different scan rates and the corresponding data is shown in figure S2. The peak current increases with scan rate⁵² while the anodic peak shift towards more positive side and cathodic peak shift towards the more negative side. Increase in the peak separation (shift in anodic and cathodic peaks) (Figure 10a) with scan rate is indication of

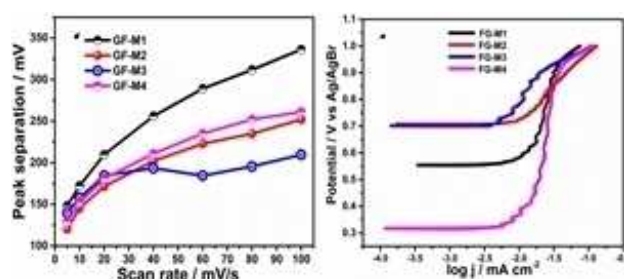


Figure 10. (a) A plot of Peak separation versus scan rate. Increasing peak separation with scan rate is direct evidence for quasireversibility of bromine redox reaction on FG modified GC electrode. (b) Tafel plot of bromide ion oxidation over difference FG modified GC electrode (b).

quasi-reversible reaction⁵⁵ which is observed in the case of CNT^{52,53} and HGs.⁴⁵ The kinetic parameters such as rate constant (k) for a quasi-reversible reaction can be calculated by

using the Nicholson's method^[52,55] and exchange current density (j_0) from Tafel plots (Figure 11b). The kinetic data obtained from CV and Tafel plots is given in table 5. The Rate constant (9.92×10^{-4} cm/s)

Table 5. Kinetic parameters obtained from CV (Figure S1) and Tafel plots (figure 11b). Exchange current density is calculated from Tafel plots, the rate constant by using the Nicholson's method and the diffusion coefficients are from Randles – Sevcik equation.

	Exchange current density (j_0) (mA/cm ²)	Rate constant K_0 (cm/s)	Diffusion coefficient	
			Br ⁻	Br ₂
FG-M1	3.71×10^{-3}	5.03×10^{-4}	6.02×10^{-6}	1.85×10^{-6}
FG-M2	52.42×10^{-3}	9.92×10^{-4}	1.19×10^{-5}	1.11×10^{-5}
FG-M3	29.72×10^{-3}	8.17×10^{-4}	1.03×10^{-5}	6.15×10^{-6}
FG-M4	1.45×10^{-3}	2.61×10^{-4}	1.00×10^{-5}	9.10×10^{-6}

10^{-4} cm/s) and exchange current density (52.42×10^{-3} mA/cm²) values are high for the materials with high fluorine content (i.e FG-M2) than that of other FGs. The variation in the kinetic parameters underscore the importance of modulating the functional groups to improve the bromine redox kinetics. In order to test the stability of FGs under aggressive bromine environment, CV studies are carried out for 50 cycles and the data is shown in Figure S3. As shown in the figure there is no much change either in peak position or peak current even after 50 cycles to indicate that irrespective of the degree of functionalization FGs are highly stable even in an aggressive bromine environment. These results show the stability of the functionalized graphene samples, in tune with our earlier studies on similar samples for their prolonged electrochemical application.^[45]

Conclusions

We have synthesized FGs with different degree of functionalization through EE of graphite in KF aqueous solution having different concentration. The physical characterizations confirm the formation of FGs with different degree of functionalization and the reasons for the variation in the functionalization are explained with appropriate data from electrochemical characterization. The as-synthesized FGs are used as electrode materials for $2\text{Br}^-/\text{Br}_2$ redox reaction. The CV results suggest that the $2\text{Br}^-/\text{Br}_2$ redox reaction is highly reversible and the peak currents are also higher on highly fluorinated graphene. Exchange current density and rate constant values are 52.42×10^{-3} mA/cm² and 9.92×10^{-4} cm/s respectively for FG-M2 (highly fluorinated FG) which shows better performance as bromine electrocatalyst. Bromine and bromide are selectively adsorbed on the surface of the FGs depending on the degree of functionalization. Irrespective of nature and degree of functionalization all HG are highly stable in the bromine environment. We strongly believe that this simple method may

facilitate participation of RFB for the large-scale energy storage from renewable energy sources such as solar and wind.

Supporting Information Summary

A detailed information about experimental procedure and characterization techniques, a table of different methods employed to synthesize fluorographene, FTIR spectrum of FGs and Cyclic Voltammograms of bromine redox reaction at different scan rate and with cycle number are given in the supporting information.

Acknowledgements

Munaiah.Yeddala and Ragupathi Pitchai acknowledges University Grants Commission (UGC), India and Department of Science and Technology (DST), India for financial support (Grant number EMR/2016/000525).

Conflict of Interest

The authors declare no conflict of interest.

Keywords: Bromine kinetics; Electrochemical exfoliation; Electrocatalyst; Graphene, Tunable functionalization; Flow battery.

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Submitted: June 7, 2019

Accepted: September 30, 2019